Constructive geometry / root iterations / global convergence

Geometric Root Calculators

A ninth-order nested-radical support and real-degree enclosures in the Pandrosion rectangle

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https://github.com/ivan-fr/pandrosion

Abstract. We study ruler-and-compass iterations for the positive root $X^{1/p}$ that share one rectangle, one power stage and one readout, and differ only in a *support line*. Every support approximates the same target function $\kappa_*(t) = (t-1)/(t^{1/p}-1)$ of the residual t, and the contact order of the approximation fixes the order of the iteration. Besides the classical supports of orders one to four we treat two radical supports. A decentered arc gives a globally monotone fifth-order iteration. A support built from two successive mean proportionals has local order *exactly nine for every real* $p > 2$, with an explicit positive error constant equal to $7/648$ for cube roots, and converges monotonically from every positive start. For integer $p \geq 3$ one nested-radical step costs at most $10p + 59$ elementary traces against $10p + 9$ for the decentered arc; measured by the efficiency index the decentered arc is preferable for $p \leq 12$ and the nested support for $p \geq 13$. We then separate what is a finite construction from what is only a scalar statement: the fast iterations are finite constructions for rational degrees, whereas for irrational p a single step is not. For arbitrary real $p \neq 0$ a power-free subdivision with one final Padé readout encloses the root with relative width at most $z^5/256$ in at most $15n + 63$ traces after n subdivisions (for $X > 1$, $p \geq 1$). Global convergence and the local orders with their constants, for both radical supports and every real $p > 2$, are checked in Lean, as are the degree, approximation and efficiency statements; the drawing inventories are written protocols.

1 Introduction

A geometric iteration must realize its arithmetic by intersections, parallels and length transfers. The Pandrosion rectangle does this for pth roots: a chain of similar triangles produces the power s^p of the current estimate, a line through a prepared point transports the residual, and a horizontal reads the next estimate on the same rail. The only part that changes from one method to another is the *support*, a line through the corner A whose slope encodes a correction factor.

Results.

1. **One target function** (Section 2). All supports approximate $\kappa_*(t) = (t-1)/(t^{1/p}-1)$; contact of order ω at $t = 1$ gives an iteration of order $\omega + 1$ with an explicit error constant (Lemma 2.1). Newton, Halley and the [1/1] Padé support are the cases $\omega = 1, 2, 3$.
2. **Decentered arc** (Section 3). One moving arc and one join give $\omega = 4$: a fifth-order iteration, monotone and convergent from every positive start for real $p > 2$ (Theorem 3.1).
3. **Nested radicals** (Section 4). Two successive mean proportionals give $\omega = 8$. The iteration has order exactly nine for every real $p > 2$, with the closed-form constant of Theorem 4.1; it is globally monotone (Theorem 4.4).
4. **Work** (Section 5). A complete per-step inventory of the nested support (Proposition 5.1), a comparison of six supports at a common state, and an efficiency-index comparison with a crossover at $p = 13$ (Proposition 5.2).
5. **Degrees** (Sections 4.5 and 6). A classification of what is constructible in finitely many traces per step, summarized in Table 1, and a direct enclosure for every real $p \neq 0$ (Propositions 6.1 and 6.2).

[1]Computational assistance: AI was used for derivations, proof development, figures and manuscript preparation. The author is responsible for the claims and their attribution. Section 8 states, claim by claim, which results are formal proofs, written proofs, computer-algebra identities or numerical checks.

Table 1: Which calculator applies to which degree. "Finite step" means that one iteration step is a finite ruler-and-compass construction from the current state.

Degree p	Fast rectangle iterations (orders 2–9)	Direct subdivision (Section 6)
Integer $p \geq 3$	Finite step; $10p+9$ or $10p+59$ traces	Yes
Rational $p = \mu/\nu$	Finite step, as degree μ applied to X^ν (Proposition 4.5)	Yes
Irrational p	Step not finite: u^p is itself a real power. Available only through a rational approximation of p (Proposition 4.7)	Yes; this is the recommended route

What "construct" means here. As in Pandrosion's recursive geometry, nothing in this paper constructs $X^{1/p}$ exactly in finitely many operations; $\sqrt[3]{2}$ is not constructible. A calculator is a rule whose every step is a finite construction and whose successive outputs converge to the root, with a stated rate or a stated enclosure. The distinction that matters is therefore not exact versus approximate, but whether *each step* is finite. Scalar convergence theorems are stated for real p; drawing inventories are stated only where a step is a finite construction.

Relation to earlier versions. The classical supports and the decentered arc were introduced in the earlier preprint [22]; they are restated here with complete proofs so that the comparison is self-contained. New in the present line of versions are the nested-radical support and its inventory. New in this revision are the contact lemma, the general ninth-order constant (which closes a question left open in V27), the efficiency-index comparison, and the degree classification of Table 1 with the rational and irrational cases treated separately. Limits of the results are collected once, in Section 9.

2 Coordinates, arithmetic and the common support equation

2.1 States and notation

For $X > 0$ and $p \neq 0$ let $r = X^{1/p}$ be the root and $\alpha = 1/r$ its reciprocal. The rectangle carries a *reciprocal state* $s > 0$ approaching α; the *direct state* is $u = 1/s$. The *residual* is

$$t = Xs^p = X/u^p, \qquad t = 1 \iff s = \alpha.$$

The logarithmic error is $e = \log(s/\alpha)$, so that $t = e^{pe}$. An iteration is written $s_+ = s\,\Phi(t)$. Table 2 fixes the main symbols.

Table 2: Notation.

$X,\ p,\ r,\ \alpha$	input, degree, root $X^{1/p}$, reciprocal root $X^{-1/p}$
$s,\ u,\ t,\ e$	reciprocal state, direct state $1/s$, residual Xs^p, log error $\log(s/\alpha)$
$h,\ \xi,\ x$	$h = 1/p$; $\xi = t-1$; $x = \log t$
$W, H;\ A, B, C, O$	rectangle width and height; its corners (Section 2.2)
$R(q), L(q), \Omega_f$	right-rail point, left-rail point, projection centre on the top line
M, E, P, T	prepared point, power endpoint $R(s^p)$, state point $R(s)$, intersection on the support
$\kappa,\ \kappa_*$	normalized support slope; exact slope $(t-1)/(t^h-1)$
$\mathsf{j}, \mathsf{p}, \mathsf{c}$	one join, one constructed parallel ($= 2\mathsf{j} + 3\mathsf{c}$), one compass arc
$q_5,\ q_9$	radicals of the decentered and nested supports
a, b	generic positive rail values; in Section 6, the moving endpoints
m, λ, z	(Section 6 only) mean proportional $\sqrt{ab}$, weight, gap $b/a - 1$
ω	order of contact of a support with κ_* at $t = 1$
μ, ν	numerator and denominator of a rational degree $p = \mu/\nu$

2.2 Rectangle and line arithmetic

In the rectangle $C = (0,0)$, $B = (W,0)$, $O = (0,H)$, $A = (W,H)$, with $W, H > 0$, define

$$R(q) = (W, H(1-q)), \qquad L(q) = (0, H(1-q)), \qquad \Omega_f = \Big(\frac{Wf}{f-1}, H\Big) \quad (f \neq 1). \tag{2.1}$$

Line incidences perform multiplication and division:

$$\Omega_f R(q) \cap OC = L(fq), \qquad \Omega_f L(b) \cap AB = R(b/f). \tag{2.2}$$

Substituting the coordinates into the line equations proves both identities. Unit factors and zero values reuse existing points instead of a centre at infinity.

Triangle power stage. Let $B_j = (Ws^j, H(1-s^j)) \in OB$ and $L_j = L(s^j)$. Horizontals transfer between these points and the rails. The parallel to L_1B through B_j meets the left rail at L_{j+1}, since its ordinate there is $H(1-s^j) + H(1-s)s^j = H(1-s^{j+1})$. For integer p, repetition gives $E = R(s^p)$ with $2p-1$ parallels and the join L_1B.

Products. For positive rail values a, b with $b \neq 1$, draw $V = CR(b) \cap OA = \Omega_{1/b}$ and project $R(a)$ successively through $\Omega_{1/2}$, V, $\Omega_{1/2}$, $\Omega_{1/4}$. By (2.2) the successive values are $L(a/2)$, $R(ab/2)$, $L(ab/4)$, $R(ab)$: a general product costs five joins. Division is symmetric: project $R(a)$ through V to $L(a/b)$ and return through $\Omega_{1/2}, \Omega_{1/4}, \Omega_{1/2}$, with values $R(2a/b)$, $L(a/(2b))$, $R(a/b)$; again five joins. Multiplication by a *prepared* constant $d > 0$ uses the two centres $\Omega_{d/[2(1+d)]}$ and $\Omega_{1/[2(1+d)]}$ and costs two joins.

2.3 The common support equation and its target

Prepare $M = (0, H(1-X^{-1}))$. With $P = R(s)$ and $E = R(s^p)$, the line ME has slope $\sigma = H(1-t)/(WX)$. Let a *support* be a line through A of slope $H\kappa/(WX)$. The parallel to ME through P meets the support at T, and the horizontal through T meets the right rail at $P_+ = R(s_+)$ with

$$s_+ = s\,\frac{\kappa}{\kappa - 1 + t}. \tag{2.3}$$

Indeed the two lines are $y - H = \frac{H\kappa}{WX}(x - W)$ and $y - H(1-s) = \sigma(x - W)$; eliminating x gives $H - y = Hs\kappa/(\kappa - 1 + t)$. The update is exact, $s_+ = \alpha$, precisely when $\kappa/(\kappa - 1 + t) = t^{-h}$, that is when

$$\kappa = \kappa_*(t) := \frac{t-1}{t^h - 1} \qquad (\kappa_*(1) = p). \tag{2.4}$$

The function κ_* is not constructible from t; every support below is a constructible approximation of it.

Lemma 2.1 (Contact lemma; limit form formalized in Lean)**.** *Let $\kappa(t)$ be analytic and positive near $t = 1$ with $\kappa(t) - \kappa_*(t) = c\,\xi^\omega + O(\xi^{\omega+1})$, $\xi = t - 1$, $\omega \geq 1$. Then the iteration (2.3) satisfies*

$$e_+ = c\,p^{\omega-1} e^{\,\omega+1} + O(e^{\,\omega+2}).$$

Proof. From (2.3), $e_+ = e + \log\kappa - \log(\kappa - 1 + t)$, and this vanishes for $\kappa = \kappa_*$. Since $\partial_\kappa[\log\kappa - \log(\kappa - 1 + t)] = (t-1)/[\kappa(\kappa - 1 + t)]$, which equals $\xi/p^2 + O(\xi^2)$ near $\kappa = p$, we get $e_+ = c\,\xi^{\omega+1}/p^2 + O(\xi^{\omega+2})$. Finally $\xi = e^{pe} - 1 = pe + O(e^2)$. □

AK is the value of κ_* at $t = 1$, AD its first-order Taylor polynomial and projective AD its [1/1] Padé approximant; these three are classical (Newton under inversion, Halley, and a Padé correction) and are documented in Appendix A. The linear MA construction uses the support MB, which does not pass through A, and is kept as a baseline in Appendix B.

Table 3: The supports as approximations of κ_*. Leading constants are for the reciprocal-state log error e; for odd orders they are unchanged in the direct state.

Support	$\kappa(t)$	contact ω	order	leading constant of e_+
AK (Newton)	p	1	2	$-(p-1)/2$
AD (Halley)	$\frac{1}{2}[p+1+(p-1)t]$	2	3	$(p^2-1)/12$
Projective AD	$\dfrac{2p[(2p-1)t+p+1]}{(p+1)t+5p-1}$	3	4	$-(p^2-1)(2p-1)/72$
Decentered AD	$1+(p-1)q_5(t)$	4	5	$(p-2)(p^2-1)(3p+1)/1440$
Nested radicals	$1+(p-1)q_9(t)$	8	9	$c_9(p)$, Theorem 4.1

2.4 Work convention

A *trace* is one join or one compass arc. A constructed parallel expands to $\mathsf{p} = 2\mathsf{j}+3\mathsf{c}$, five traces. Intersections of drawn objects are free. *Fixed preparation*, meaning every object that depends only on p, X, W, H and not on the current state, is not charged, nor are decisions (comparisons of two constructed lengths) or the single final inversion that converts s into $u = 1/s$. Section 6 charges its input and output inversions explicitly. All counts are upper bounds for the specified protocols; a trace count is neither a time nor an energy.

The name Pandrosion refers to the addressee of Book III of Pappus' *Collection* [1]. Our rectangle follows a modern reconstruction [2]; the accelerated supports are modern and are not attributed to the ancient text.

3 A decentered arc with global fifth-order convergence

3.1 Formula and construction

For real $p > 2$ put

$$S(t) = (p-2)(t^2+1) + (10p+4)t, \qquad q_5(t) = \sqrt{\frac{S(t)}{12p}}, \qquad \mathrm{T}_{\mathrm{arc}}(s) = s\,\frac{1+(p-1)q_5(t)}{t+(p-1)q_5(t)}, \tag{3.1}$$

that is, the support $\kappa = 1+(p-1)q_5(t)$ in (2.3). The radicand is positive for $t > 0$.

For integer $p \geq 3$ the radical is produced by one arc. Put

$$\eta = (p-1)\sqrt{\frac{p-2}{12p}}, \qquad \zeta = \frac{5p+2}{p-2} > 1, \qquad d_0 = \frac{W}{\eta}, \qquad \delta = \frac{H}{X}\sqrt{\zeta^2-1},$$

and prepare

$$G_0 = \Big(W,\ H + \frac{H\zeta}{X}\Big), \qquad O_{\mathrm{arc}} = \Big(W - d_0 + \delta,\ H - \frac{H}{X\eta}\Big), \qquad \text{the vertical } x = W - d_0.$$

Draw the arc centred at O_{arc} with radius $EG_0 = (H/X)(t+\zeta)$ and take its lower intersection D_{arc} with the prepared vertical. Its distance below the centre is

$$\sqrt{EG_0^2 - \delta^2} = \frac{H}{X}\sqrt{t^2+2\zeta t+1},$$

so the slope of AD_{arc} is $\frac{H}{WX}\big[1+\eta\sqrt{t^2+2\zeta t+1}\big] = \frac{H}{WX}\big[1+(p-1)q_5(t)\big]$, because $(p-2)\cdot 2\zeta = 10p+4$. All preparation parameters use rational operations and square roots of positive quantities. Centered AD and decentered AD use the same mobile operations, one arc and one join; only the fixed preparation differs. This is the precise sense in which order five is obtained at the mobile cost of order three.

3.2 Global monotone convergence

Theorem 3.1 (Formalized in Lean)**.** *For real $p > 2$, $X > 0$ and every $s_0 > 0$, the iteration $s_{n+1} = \mathrm{T}_{\mathrm{arc}}(s_n)$ converges to α monotonically without crossing it, and*

$$e_+ = \frac{(p-2)(p^2-1)(3p+1)}{1440}\,e^5 + O(e^7). \tag{3.2}$$

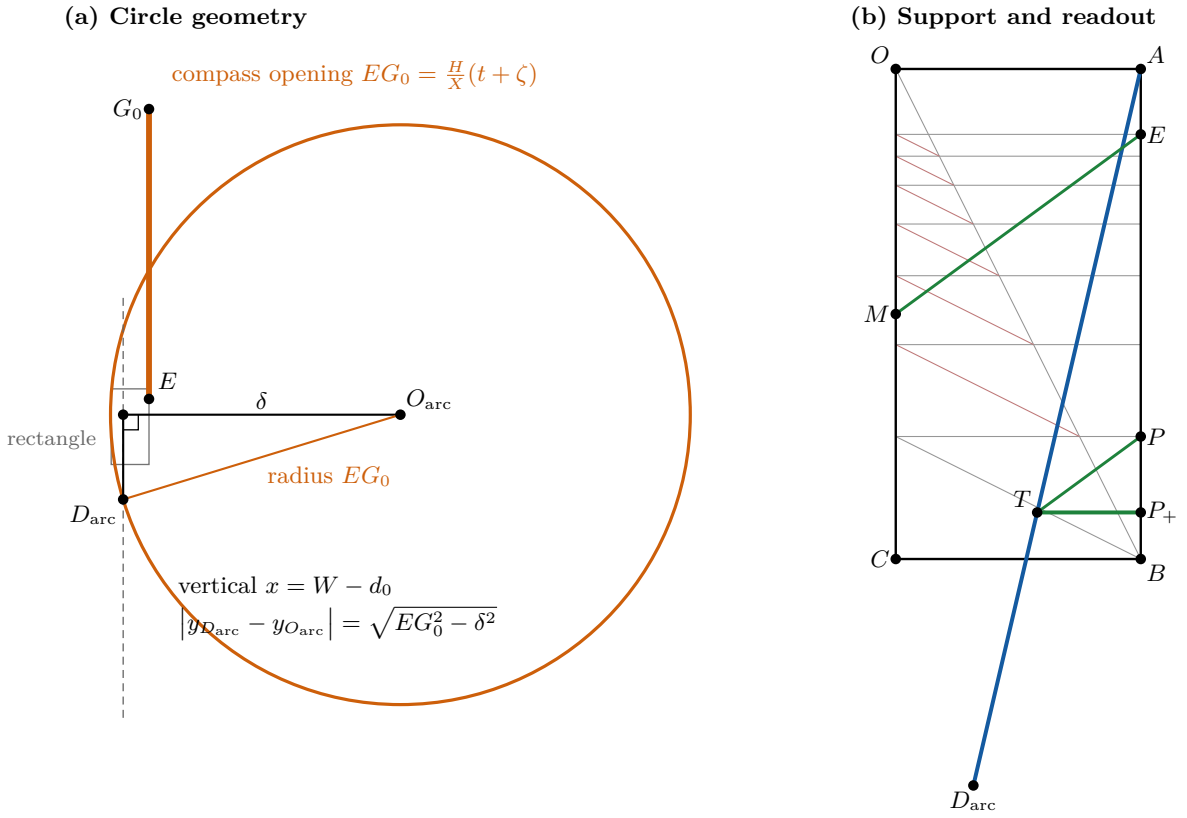


Figure 1: The decentered-arc device for $p = 7$, $X = W = 2$, $H = 4$, $s = 3/4$, drawn from the exact coordinates. (a) The whole circle. The compass opening EG_0 (thick orange segment) is carried to the prepared centre O_{arc}; the circle meets the prepared vertical at D_{arc}, and the right triangle with horizontal leg δ shows why its vertical leg is a square root. The small grey box is the rectangle at the same scale. (b) The rectangle enlarged: triangle chain (grey), transport lines (green), support AD_{arc} (blue) and the readout P_+. Each panel has equal axis scales.

Proof. Write $g = (p-1)q_5$. Since $S(1/t) = S(t)/t^2$ we have $q_5(1/t) = q_5(t)/t$, hence $\Phi(1/t) = 1/\Phi(t)$ for $\Phi = (1+g)/(t+g)$: the log-error map $e \mapsto e_+$ is odd. Put

$$\Lambda(t) = \frac{\log t}{p} + \log(1+g(t)) - \log(t+g(t)),$$

so that $e_+ = \Lambda(t)$, and define the polynomials

$$\begin{aligned} N_1(t) &= (p-1)(t^2+1) + (10p+2)t, \\ N_2(t) &= (t+1)\big(6pt - (t-1)^2\big), \\ N_3(t) &= (p^2-5p-2)(t-1)^2 + 12p(3p+1)t. \end{aligned}$$

Differentiation and factorization give the two identities

$$\frac{d\Lambda}{d\log t} = \frac{(p-2)(p-1)^2\left(q_5 N_1 - N_2\right)}{12p^2\, g(1+g)(t+g)}, \tag{3.3}$$

$$q_5^2 N_1^2 - N_2^2 = \frac{p+1}{12p}\,(t-1)^4\, N_3. \tag{3.4}$$

Here $N_1 > 0$. If $N_2 \le 0$ then $q_5 N_1 - N_2 > 0$. Otherwise $(t-1)^2 < 6pt$, and $N_3 > 0$: directly when $p^2 - 5p - 2 \ge 0$, and otherwise from the rearrangement

$$N_3 = 6p^2(p+1)t + (p^2-5p-2)\big((t-1)^2 - 6pt\big),$$

whose two terms are positive. Then (3.4) gives $q_5 N_1 > N_2$ for $t \ne 1$. Hence Λ is strictly increasing with $\Lambda(1) = 0$: the new log error has the sign of the old one. Meanwhile $\Phi(t) - 1 = (1-t)/(t+g)$ moves the state toward α. The orbit is monotone and bounded between s_0 and α; its positive limit is a fixed point by continuity, and $\Phi(t) = 1$ forces $t = 1$.

Near $t = 1$, rationalize $q_5N_1 - N_2$ with (3.4). At $t = 1$ one has $q_5 = 1$, $g = p - 1$, $N_1 = N_2 = 12p$ and $N_3 = 12p(3p + 1)$, so

$$\lim_{t\to 1} \frac{d\Lambda/d\log t}{(t-1)^4} = \frac{(p-2)(p^2-1)(3p+1)}{288\,p^5}.$$

Substituting $t = e^{pe}$ and integrating yields the coefficient in (3.2); oddness removes the e^6 term. □

At $p = 3$ the constant is $1/18$. At $p = 2$ the scalar formula is an exact square-root correction, but the displayed preparation is singular and is replaced by a single mean proportional.

4 A nested-radical support of order nine

The next support uses two successive mean proportionals, the output of the first entering the diameter of the second. In this section it is convenient to follow the direct state $u = 1/s$ with target $r = X^{1/p}$ and log error $\hat{e} = \log(u/r) = -e$. The residual is unchanged, $t = X/u^p = Xs^p$, so the rail encoding of Section 2 is unchanged.

4.1 Formula, domain and rectangle support

For $p > 2$ write $h = 1/p$ and define

$$a_9 = \frac{1-2h}{12}, \qquad b_9 = \frac{(h+1)(h+3)}{30}, \qquad c_9 = \frac{121 - 185h - 67h^2 - h^3}{840(h+3)}, \tag{4.1}$$

$$w = \frac{(t-1)^2}{t}, \qquad k_9 = \sqrt{1 + \frac{b_9 w}{1 + c_9 w}}, \qquad q_9 = \sqrt{t\Big(1 + \frac{a_9 w}{k_9}\Big)}. \tag{4.2}$$

All square roots are positive branches. The iteration is

$$\Gamma_p(t) = \frac{t + (p-1)q_9(t)}{1 + (p-1)q_9(t)}, \qquad F_{p,X}(u) = u\,\Gamma_p\big(X/u^p\big), \tag{4.3}$$

which is the support $\kappa = 1 + (p-1)q_9$ in (2.3), read in the direct state.

For $0 < h < 1/2$ the three coefficients are positive: the numerator of c_9 is decreasing on this interval and equals $93/8$ at $h = 1/2$. Thus every division and radical in (4.2) is defined for every $t > 0$. Moreover

$$w(1/t) = w(t), \qquad q_9(1/t) = q_9(t)/t, \qquad \Gamma_p(1/t) = 1/\Gamma_p(t), \tag{4.4}$$

so the log-error map $\hat{e} \mapsto \hat{e}_+$ is odd.

In the rectangle, one support point is $D_9 = \big(0,\ H - \frac{H}{X}(1 + (p-1)q_9)\big)$. The parallel to ME through $P = R(s)$ meets AD_9, and the horizontal readout gives $s_+ = s\,(1 + (p-1)q_9)/(t + (p-1)q_9)$, i.e. $1/s_+ = F_{p,X}(u)$. After t is obtained, the arithmetic consists of rational operations and two mean proportionals, $k_9 = \sqrt{1 \cdot (1 + b_9 w/(1 + c_9 w))}$ and $q_9 = \sqrt{t \cdot (1 + a_9 w/k_9)}$.

4.2 First complete step for the cube root of two

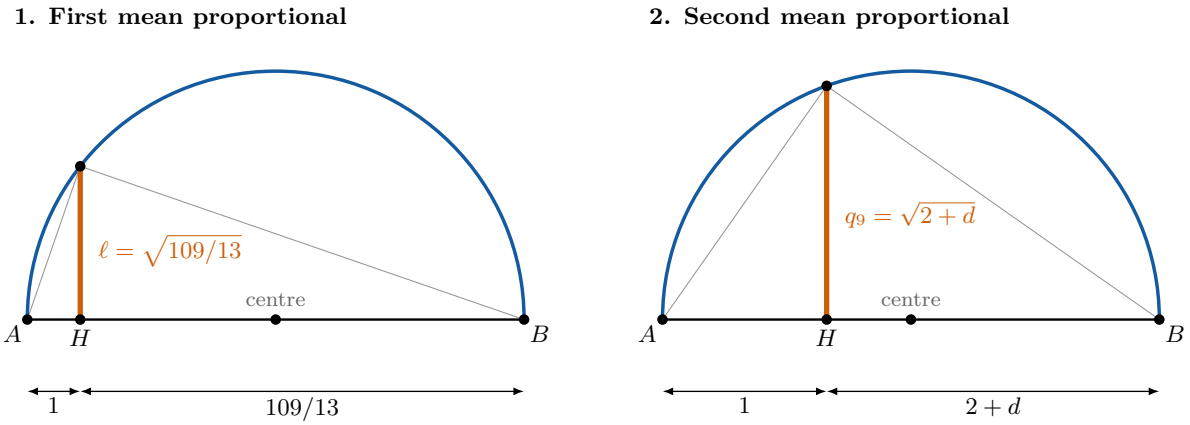
At $p = 3$ the support simplifies to

$$q_9(t) = \sqrt{t + \frac{(t-1)^2}{108}\sqrt{\frac{t^2 + 52t + 1}{t^2 + 4t + 1}}}, \qquad u_+ = u\,\frac{t + 2q_9(t)}{1 + 2q_9(t)}. \tag{4.5}$$

Here is the first step for $X = 2$, $u_0 = 1$. The auxiliary diagrams have their own local point names. The unknown cube root is never used to place a point.

1. *Rational preparation.* The residual is $t = 2$. Prepare $109/13$ and $1/108$ by subdivisions and similar triangles.

2. *First semicircle.* Place adjacent segments of lengths 1 and 109/13, draw the semicircle on their sum and the perpendicular at the junction. Its height is $\ell = \sqrt{109/13} = 2.895620034572\ldots$ (This cubic auxiliary is not the general k_9.)
3. *Second diameter.* A similar-triangle division gives $d = \ell/108 = 0.026811296616\ldots$; add it to two units.
4. *Second semicircle.* With adjacent lengths 1 and $2 + d$ the perpendicular height is $q_9 = \sqrt{2+d} = 1.423661229583\ldots$
5. *Move the support.* Choose $W = 6$, $H = 4$, so $M = (0, 2)$. Here $s_0 = 1$, hence $P_0 = E = B$. Place $D_9 = (0,\, 2 - 4q_9)$ below M.
6. *Intersect and report.* Draw AD_9 and MB; they meet at T_9. The horizontal through T_9 meets AB at P_1, and the rectangle reports $s_1 = AP_1/AB$.
7. *Invert.* A similar-triangle construction produces $u_1 = AB/AP_1$.



Between the two: prepare $d = \ell/108$ by similar triangles, then transfer $2 + d$ to the second diameter.

Figure 2: The two successive mean proportionals for $X = 2$, $u_0 = 1$. Each panel has equal axis scales and its own magnification. The orange perpendiculars are the constructed lengths; the first height supplies the rationally scaled addition d used by the second diameter.

The incidences give

$$\begin{aligned} T_9 &= (3.524406494288\ldots,\ 0.825197835237\ldots), \\ s_1 &= 0.79370054119068191329\ldots, \\ u_1 &= 1.25992102575592908435636220628 1\ldots, \end{aligned}$$

with relative error $u_1/r - 1 \approx -1.9159092613 \times 10^{-8}$. These digits describe exact geometric incidences evaluated numerically; they are not readable to that precision from a manual drawing. Later steps recompute $t = 2/u^3$: the constants 109 and 13 belong to this first step only. The seven grouped instructions above, like the twelve screens of the companion interactive display, are pedagogical steps and not primitive trace counts; those are in Section 5.

4.3 Local order: exactly nine for every real $p > 2$

The exact support corresponds to $q_* = (\kappa_* - 1)/(p-1) = (t - t^h)/[(p-1)(t^h - 1)]$, used only in the analysis. With $x = \log t$ and quotients interpreted continuously at zero,

$$Y := \frac{q_*^2}{t} = \left[\frac{\sinh((1-h)x/2)\,/\,((1-h)x/2)}{\sinh(hx/2)\,/\,(hx/2)}\right]^2, \qquad w = 2(\cosh x - 1).$$

Both sides are even in x, so Y is a power series in w:

$$\begin{aligned} Y &= 1 + a_9 w + A_2 w^2 + A_3 w^3 + O(w^4), \\ A_2 &= -\frac{(1-2h)(h+1)(h+3)}{720}, \qquad A_3 = \frac{(1-2h)(h+1)(2h^3+8h^2+13h+31)}{60480}, \end{aligned} \tag{4.6}$$

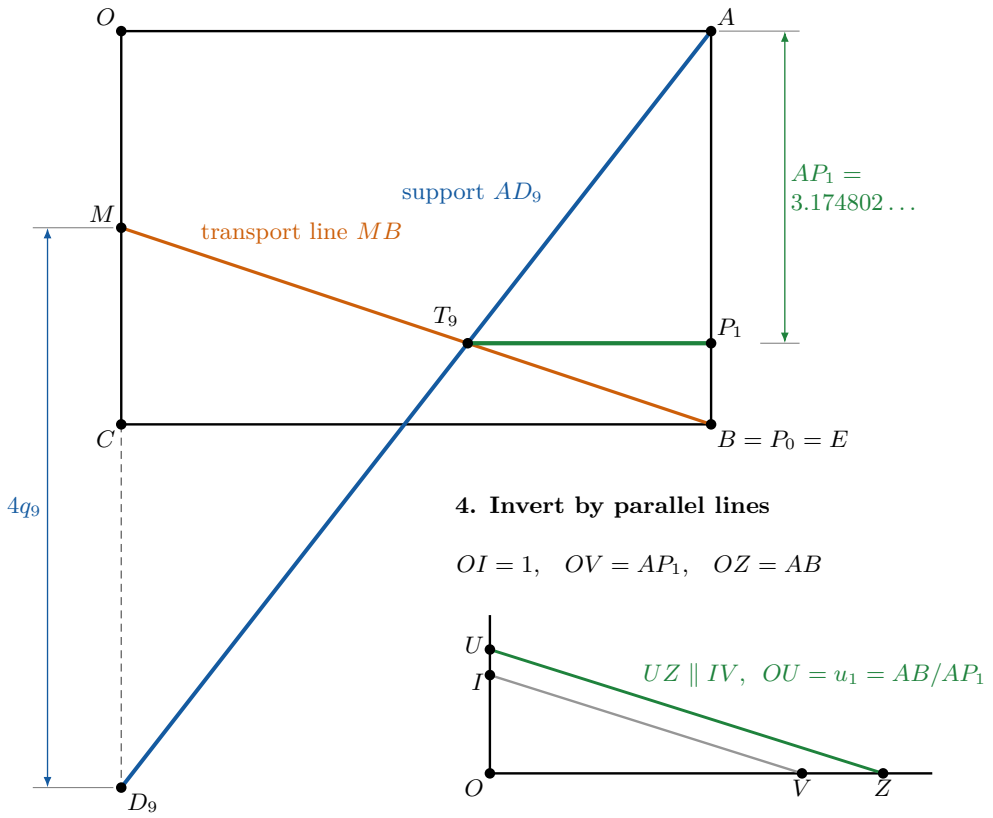


Figure 3: The modified Pandrosion support for the same first step, at equal axis scales. The support AD_9 (blue) meets the transport line (orange) at T_9, and the horizontal (green) reads P_1. Because $s_0 = 1$, $P_0 = E = B$ and the transport line is MB itself. The extension below C is part of the construction. Lower right: the reciprocal reading $u_1 = AB/AP_1$ by parallel lines.

while the proposed support has

$$\frac{q_9^2}{t} = 1 + a_9 w - \frac{a_9 b_9}{2}\, w^2 + a_9\Big(\frac{b_9 c_9}{2} + \frac{3b_9^2}{8}\Big) w^3 + O(w^4). \tag{4.7}$$

The coefficients (4.1) are exactly those that make (4.7) agree with (4.6) through w^3. The first unmatched term is explicit.

Theorem 4.1 (Order nine with explicit constant; formalized in Lean)**.** *For every real $p > 2$ and $X > 0$, the iteration $u_+ = F_{p,X}(u)$ satisfies*

$$\hat{e}_+ = c_9(p)\, \hat{e}^{\,9} + O(\hat{e}^{\,11}), \qquad c_9(p) = \frac{(p^2-1)(p-2)\, N_9(p)}{2^{11}\, 3^4\, 5^3\, 7^2\, (3p+1)}, \tag{4.8}$$

$$N_9(p) = 21311p^6 + 1650p^5 - 21679p^4 - 15532p^3 - 4411p^2 - 526p - 29.$$

Moreover $c_9(p) > 0$ for every $p > 2$, so the local order is exactly nine. In particular $c_9(3) = 7/648$.

Proof. Continuing the two expansions one term further gives

$$\frac{q_9^2}{t} - Y = D_4\, w^4 + O(w^5), \qquad D_4 = \frac{(1-2h)(1+h)\, n_9(h)}{2^{10}\, 3^4\, 5^3\, 7^2\, (h+3)}, \tag{4.9}$$

with $n_9(h) = 21311 + 1650h - 21679h^2 - 15532h^3 - 4411h^4 - 526h^5 - 29h^6 = h^6 N_9(1/h)$. This is an identity between rational functions of h, obtained by exact series expansion. (In the Lean proof it is obtained without Taylor series: $h^2t^4(t-t^h)^2 - (1-h)^2 t(t^h-1)^2\, t^4 P_4(w)$, with P_4 the quartic part of (4.6), is a sum of 27 exponentials $c_j e^{\lambda_j x}$ whose moments $\sum_j c_j \lambda_j^n$ vanish for $n < 12$, hence it is $O(x^{12})$.) Since $w = \xi^2/t$ and

$q_9 + q_* \to 2$ as $t \to 1$, we get $q_9 - q_* = \frac{1}{2}D_4\xi^8 + O(\xi^9)$ and therefore $\kappa - \kappa_* = \frac{1}{2}(p-1)D_4\,\xi^8 + O(\xi^9)$. Lemma 2.1 with $\omega = 8$ gives $e_+ = \frac{1}{2}(p-1)p^7 D_4\,e^9 + O(e^{10})$. Substituting $h = 1/p$,

$$\tfrac{1}{2}(p-1)p^7 D_4 = \frac{(p-1)(p-2)(p+1)N_9(p)}{2^{11}3^4 5^3 7^2(3p+1)} = c_9(p).$$

By (4.4) the map is odd, so the same expansion holds for $\hat{e} = -e$ and the next term is of order eleven. For positivity,

$$N_9(2+\rho) = 21311\rho^6 + 257382\rho^5 + 1273481\rho^4 + 3286796\rho^3 + 4628741\rho^2 + 3325430\rho + 926859,$$

which has only positive coefficients; hence $N_9(p) > 0$ and $c_9(p) > 0$ for $p > 2$. Finally $N_9(3) = 13720000$ gives $c_9(3) = 8 \cdot 13720000/(1016064000 \cdot 10) = 7/648$. □

Remark 4.2 (Scale of the constant). $c_9(p)$ grows like p^8: for instance $c_9(4) \approx 0.187$, $c_9(7) \approx 26.6$ and $c_9(20) \approx 1.58 \times 10^5$. This is the natural scale, not a deterioration. In the residual logarithm $x = \log t = -p\hat{e}$ the law reads $x_+ = c_9(p)p^{-8}x^9 + \dots$, and $c_9(p)p^{-8} \to 21311/3048192000 \approx 7.0 \times 10^{-6}$ as $p \to \infty$. The same rescaling applied to (3.2) gives the limit $1/480$ for the decentered arc. The asymptotic regime is $|x| \lesssim 1$, that is $|\hat{e}| \lesssim 1/p$, for both.

For the cubic specialization a short positive certificate proves the constant independently of series manipulations; it was the first version formalized in Lean. Let $v = t^{1/3} > 0$ and set

$$\begin{aligned} D_v &= v^6 + 4v^3 + 1, \\ H_v &= v^8 + 10v^7 + 55v^6 + 268v^5 + 466v^4 + 268v^3 + 55v^2 + 10v + 1, \\ \Theta_v &= (v^2+v+1)^2\sqrt{(v^6+52v^3+1)/D_v}. \end{aligned}$$

The exact identity

$$(v^2+v+1)^4(v^6+52v^3+1) - 729v^4 D_v = (v-1)^6 H_v \tag{4.10}$$

implies $\Theta_v \geq 27v^2$ and gives

$$q_9^2 - \Big(\frac{v^2+v}{2}\Big)^2 = \frac{(v-1)^8 H_v}{108\,D_v\,(\Theta_v + 27v^2)}, \tag{4.11}$$

where every denominator is positive. Since $H_1 = 1134$, $q_9 - \frac{1}{2}(v^2+v) = \frac{7}{432}(v-1)^8 + O((v-1)^9)$.

Theorem 4.3 (Cubic certificate; formalized in Lean). *For $X > 0$ and $r = X^{1/3}$,*

$$\lim_{u \to r,\ u \neq r} \frac{F_{3,X}(u)/r - 1}{(u/r-1)^9} = \frac{7}{648}.$$

Proof. With $y = u/r$ one has $v = 1/y$. The exact difference $\Gamma_3(v^3) - v$ equals $2(1-v)\big(q_9 - \frac{1}{2}(v^2+v)\big)/(1+2q_9)$. Substituting (4.11) and $v = 1/y$ factors $F_{3,1}(y) - 1$ as $(y-1)^9$ times a continuous positive function with value $7/648$ at one. The scaling $F_{3,r^3}(ry) = rF_{3,1}(y)$ gives the limit. □

4.4 Global convergence for every real degree greater than two

Local cancellation says nothing about distant starts. An independent sign argument covers the whole positive axis.

Theorem 4.4 (Global monotone convergence; formalized in Lean). *For every real $p > 2$, $X > 0$ and $u_0 > 0$, the iteration $u_{n+1} = F_{p,X}(u_n)$ is defined, stays positive and converges to $r = X^{1/p}$. If $u_0 \neq r$ it moves strictly toward r without crossing it.*

Proof. We give the reduction to positive polynomial certificates; the complete integer coefficients are in the Lean source, and their structure was recomputed independently for this revision (Section 8). Define

$$\mathcal{E} = 840p^2(3p+1), \qquad \mathcal{C} = 121p^3 - 185p^2 - 67p - 1, \qquad \mathcal{D} = 373p^3 + 235p^2 + 129p + 27,$$
$$\mathcal{U} = \mathcal{E} + \mathcal{C}w, \qquad \mathcal{V} = \mathcal{E} + \mathcal{D}w, \qquad \mathcal{L} = \sqrt{\mathcal{U}/\mathcal{V}}, \qquad \mathcal{R} = 1 + \frac{(p-2)w\mathcal{L}}{12p}.$$

Then $\mathcal{L} = 1/k_9$ and $q_9 = \sqrt{t\mathcal{R}}$. For $p > 2$ and $w \geq 0$ the quantities $\mathcal{U}, \mathcal{V}, \mathcal{L}, \mathcal{R}$ are positive. Put

$$\mathcal{J} = 4\mathcal{U}\mathcal{V} - p\,\mathcal{E}(\mathcal{D} - \mathcal{C})w, \qquad \mathcal{A} = 1 + \frac{(p-1)w\mathcal{L}}{12p}, \qquad \mathcal{B} = 1 - \frac{\mathcal{J}w\mathcal{L}}{24p\mathcal{U}\mathcal{V}}.$$

A two-region certificate establishes

$$0 < w \leq 30p \implies \mathcal{R}\mathcal{A}^2 > (1 + w/4)\,\mathcal{B}^2, \tag{4.12}$$

$$w \geq 30p \implies \mathcal{B} < 0. \tag{4.13}$$

The elimination uses

$$\mathcal{K} = -1728p^3\mathcal{U}\mathcal{V}^3 + 48p(p-1)(3p-5)\,\mathcal{U}^2\mathcal{V}^2w - 3p\,\mathcal{J}^2(4+w)w,$$
$$\mathcal{M} = 576p^2(3p-4)\,\mathcal{U}\mathcal{V} + 144p^2\mathcal{J}(4+w) + 4(p-2)(p-1)^2\,\mathcal{U}^2w^2.$$

There is a polynomial $\mathcal{P}(p, w)$, of degree nine in w and twenty-nine in p, with

$$(\mathcal{V}^2\mathcal{M})^2\,\mathcal{U} - \mathcal{K}^2\,\mathcal{V} = (p+1)\,w^3\,\mathcal{V}\,\mathcal{P}(p, w),$$

and it has the exact Bernstein form

$$(30p)^9\,\mathcal{P}(p, w) = \sum_{j=0}^{9} w^j(30p - w)^{9-j}\,T_j(p-2), \tag{4.14}$$

in which every T_j is a polynomial with only positive coefficients, 345 terms in all. Hence $\mathcal{P} > 0$ for $p > 2$, $0 \leq w \leq 30p$. Positive-coefficient expansions of $\mathcal{M}$ in $(p-2, w)$, and of $\mathcal{J}$ and $w^2\mathcal{J}^2 - 576p^2\mathcal{U}\mathcal{V}^3$ in $(p-2,\ w-30p)$, with 49, 24 and 112 terms respectively, provide the remaining signs. Rationalizing with $\mathcal{L}^2\mathcal{V} = \mathcal{U}$ proves (4.12) and (4.13). These are exact identities and inequalities, not sampled signs.

Since $(t+1)^2 = 4t(1 + w/4)$ and $q_9^2 = t\mathcal{R}$, the two regions give

$$2q_9\mathcal{A} - (t+1)\mathcal{B} > 0 \qquad (t > 0,\ t \neq 1). \tag{4.15}$$

Define $\Delta_p(t) = \log\Gamma_p(t) - \log(t)/p$. Exact differentiation yields

$$\Delta_p'(t) = -\frac{(p-1)(p-2)\left[2q_9\mathcal{A} - (t+1)\mathcal{B}\right]}{2p\,q_9\,[t + (p-1)q_9]\,[1 + (p-1)q_9]} < 0 \qquad (t \neq 1). \tag{4.16}$$

With $\Delta_p(1) = 0$ and $\Gamma_p(t) - 1 = (t-1)/(1 + (p-1)q_9)$ this proves

$$t > 1:\ 1 < \Gamma_p(t) < t^{1/p}, \qquad 0 < t < 1:\ t^{1/p} < \Gamma_p(t) < 1.$$

Consequently $u < F_{p,X}(u) < r$ below the root and $r < F_{p,X}(u) < u$ above it. The orbit is monotone and bounded, its limit is positive, continuity makes it a fixed point, and $\Gamma_p(t) = 1$ forces $t = 1$. □

4.5 Which degrees give a finite construction

Theorems 3.1, 4.1 and 4.4 are statements about real functions of a real parameter p. A drawing needs more: the residual $t = X/u^p$ must be *constructed* at every step. The triangle stage constructs integer powers only. This subsection says exactly what follows.

Proposition 4.5 (Rational degrees are finite constructions)**.** *Let $p = \mu/\nu$ with coprime positive integers μ, ν, and $X > 0$. Then $X^{1/p} = (X^\nu)^{1/\mu}$, and:*

(i) if $\mu \geq 3$, every integer-degree protocol of this paper applies with degree μ and input X^ν. In particular the decentered and nested iterations converge monotonically from every positive start, with orders five and nine, at no more than $10\mu + 9$ and $10\mu + 59$ traces per step (Section 5). The input X^ν is fixed preparation: $\nu - 1$ five-join products, once;

(ii) if $\mu \in \{1, 2\}$, the root X^ν or $\sqrt{X^\nu}$ is constructed exactly, by $\nu - 1$ products and at most one mean proportional.

For $p < 0$ apply the above to $|p|$ and invert the output (five joins, once).

Proof. The residual of the degree-μ problem with input X^ν is $t = X^\nu s^\mu$, built by the triangle stage with μ in place of p from the prepared point $M = (0, H(1 - X^{-\nu}))$. Theorems 3.1 and 4.4 apply with real degree $\mu > 2$ and positive input X^ν, and their fixed point is $(X^\nu)^{1/\mu}$. □

Two consequences are worth stating. First, the degree seen by the iteration is μ, not p: the degree $p = 3/2$ is handled as a cube root of X^2, inside the domain $\mu > 2$ of the unmodified formulas although $3/2 < 2$. Second, the cost is governed by μ: $p = 355/113$ costs as much per step as degree 355.

The restriction to degree greater than two is real for the *raw* formula. For $p = 3/2$, $t = 100$ one has $c_9 = -25/2376$, $1 + c_9 w = -1/32$ and $1 + b_9 w/(1 + c_9 w) = -15947/25 < 0$, so k_9 is not real; Lean checks this counterexample. As a scalar statement the restriction can be removed:

Proposition 4.6 (Scalar extension to every nonzero real degree; formalized in Lean)**.** *For every real $p \neq 0$, every $X > 0$ and every start $u_0 > 0$ there is an explicit modification of (4.3) that converges monotonically to $X^{1/p}$ without crossing it.*

Proof. Choose an integer $k \geq 0$ with $p' = 2^k |p| > 2$. For $p > 0$ put $v_0 = u_0^{1/2^k}$, iterate $v_{n+1} = F_{p',X}(v_n)$ and report $u_n = v_n^{2^k}$; for $p < 0$ put $v_0 = u_0^{-1/2^k}$ and report $u_n = v_n^{-2^k}$. The reported start is exactly u_0. Theorem 4.4 gives $v_n \to X^{1/p'}$, so $u_n \to X^{\pm 2^k/p'} = X^{1/p}$. Positive integer powers preserve inequalities and inversion reverses them together with the direction of approach. For $p > 2$ take $k = 0$. □

Proposition 4.6 is a change of variable and adds no geometric content beyond Proposition 4.5: for irrational p the step still needs the real power $v^{p'}$, which is a problem of the same nature as the root being sought. Irrational degrees can nevertheless be reached by the fast iterations through a rational approximation of p, with a controlled error:

Proposition 4.7 (Irrational degrees by rational approximation)**.** *Let $p > 0$ be real, $X > 0$, $r = X^{1/p}$, and let μ, ν be positive integers with $|\nu/\mu - 1/p| \leq \eta$. Then $r' = X^{\nu/\mu}$ satisfies*

$$\left| \frac{r'}{r} - 1 \right| \leq e^{\eta |\log X|} - 1.$$

If moreover $\tilde{u} > 0$ satisfies $|\tilde{u}/r' - 1| \leq \varepsilon_1$, then $|\tilde{u}/r - 1| \leq (1 + \varepsilon_1)\, e^{\eta |\log X|} - 1$.

Proof. $r'/r = X^{\nu/\mu - 1/p}$, so $|\log(r'/r)| \leq \eta |\log X|$, which gives the first bound since $1 - e^{-y} \leq e^y - 1$ for $y \geq 0$. For the second, $\tilde{u}/r = (\tilde{u}/r')(r'/r)$ lies between $(1 - \varepsilon_1)e^{-y}$ and $(1 + \varepsilon_1)e^y$ with $y = \eta |\log X|$, and $1 - (1 - \varepsilon_1)e^{-y} \leq (1 + \varepsilon_1)e^y - 1$. □

Remark 4.8 (Cost of the rational route). Take $p = \pi$, $X = 2$, $r = 2^{1/\pi} = 1.2468689889\ldots$ With $\mu/\nu = 22/7$, $\eta \approx 1.28 \times 10^{-4}$ and the approximation error is at most 8.9×10^{-5}; a step costs at most 229 traces (decentered) or 279 (nested), after 30 joins of preparation. With $\mu/\nu = 355/113$, $\eta \approx 2.70 \times 10^{-8}$ and the error is at most 1.9×10^{-8}, but a step costs at most 3559 or 3609 traces after 560 joins of preparation. The direct subdivision of Section 6 reaches relative width 10^{-8} for the true exponent in at most 123 traces. Since a convergent ν/μ of $1/p$ has η of order μ^{-2}, the rational route needs μ of order $\varepsilon^{-1/2}$ for accuracy ε, while its step cost is linear in μ. For irrational degrees the subdivision is therefore the recommended calculator, and the rational route is of interest only when a small μ is acceptable.

A third route would approximate the real power u^p inside each step by an inner subdivision. That is a two-level recursion whose outer iteration receives an inexact residual; its convergence would need a perturbation analysis that we have not carried out (Section 10).

4.6 Algebraic character of the correction

At $p = 3$, $t = 2$,

$$q_9^2 = 2 + \frac{\sqrt{109/13}}{108}, \qquad N_{\mathbb{Q}(\sqrt{109/13})/\mathbb{Q}}\big(q_9^2\big) = \frac{606419}{151632}.$$

Neither this norm nor $109/13$ is a rational square, so q_9 has degree four over $\mathbb{Q}$. The correction $(2 + 2q_9)/(1 + 2q_9)$ is an invertible homography of q_9 and also has degree four. Hence the cubic correction is not a rational function of t with rational coefficients, and cannot be written in this normalization with a single square root over $\mathbb{Q}$. This is an exact rational computation, not one of the Lean theorems.

The argument does not exclude generalized algebraic or Hermite–Padé approximants, transformed targets or other changes of variable. Algebraic approximants of arbitrary degree are established generalizations of Padé approximation [3], and the coefficient matching of (4.6)–(4.7) belongs to that methodology. Appendix C discusses what "the same formula up to conjugacy" can and cannot mean for the decentered arc.

5 Geometric work: inventory, comparison and efficiency

Throughout this section $p \geq 3$ is an integer (or the numerator μ of a rational degree, Proposition 4.5), the state $R(s)$ is supplied, and counts follow Section 2.4. For the five supports through A other than the nested one, the power stage uses $2p-1$ parallels and the join L_1B; transport adds the join ME and two parallels (the parallel through P and the horizontal readout). Moving AD adds one arc and one join; projective AD adds two joins.

5.1 Complete inventory of the nested support

Fixed preparation includes a_9, b_9, c_9 and $\lambda_9 = (p-1)/X$; every residual-dependent length is rebuilt at each step. Obtain $E = R(s^p)$ by the triangle stage, then $R(t)$ by the two-join fixed scale X. Stop if $t = 1$. Otherwise use

$$w = t + \frac{1}{t} - 2, \tag{5.1}$$

which needs one five-join division and two rail sums or differences instead of a separate squaring. Then construct successively

$$K = 1 + \frac{b_9 w}{1 + c_9 w}, \qquad k_9 = \sqrt{1 \cdot K}, \qquad \rho_9 = 1 + \frac{a_9 w}{k_9}, \qquad q_9 = \sqrt{t\,\rho_9}.$$

Each mean proportional uses the semicircle macro of Section 6.3, at most $2\mathsf{j} + \mathsf{p} + 3\mathsf{c}$ including diameter preparation and transfer to the rail. Scale q_9 to $\lambda_9 q_9$ in two joins; its rail length is $H\lambda_9 q_9$. One arc centred at M transfers this length downward on the left rail and gives $D_9 = (0,\ H - \frac{H}{X}(1 + (p-1)q_9))$. Join AD_9 and ME, draw the parallel to ME through P, and return horizontally.

Table 4: Per-step upper inventory of the nested support, from $R(s)$ to $R(s_+)$, integer $p \geq 3$. A sum or nonnegative difference is one rail arc, a general division five joins, a fixed scale two joins.

Stage	j	p	c
Triangle power endpoint $E = R(s^p)$	1	$2p-1$	0
Residual $t = Xs^p$	2	0	0
Width $w = t + 1/t - 2$	5	0	2
Inner argument K (two scales, one division, two sums)	9	0	2
First mean $k_9 = \sqrt{1 \cdot K}$	2	1	3
Outer argument ρ_9 (scale, division, sum)	7	0	1
Second mean $q_9 = \sqrt{t\rho_9}$	2	1	3
Scale $\lambda_9 q_9$ and transfer from M to D_9	2	0	1
Support join AD_9	1	0	0
Join ME, transport parallel, horizontal readout	1	2	0
Total	32	$2p+3$	12

Proposition 5.1 (Nested-radical work bound)**.** *For integer $p \geq 3$, $X > 0$ and a supplied state $R(s)$, the protocol above realizes the nested-radical correction with at most*

$$32\mathsf{j} + (2p+3)\mathsf{p} + 12\mathsf{c} = (4p+38)\mathsf{j} + (6p+21)\mathsf{c}, \tag{5.2}$$

that is $10p + 59$ traces per nonterminal step; 89 traces for cube roots.

Proof. The rail identities (2.2) establish every division and fixed scale. Both mean-proportional inputs are positive, and the prescribed circle intersection with the transferred length gives the positive root. The point D_9 gives $\kappa = 1 + (p-1)q_9$ in (2.3). Adding the ten rows gives the left side of (5.2), and $\mathsf{p} = 2\mathsf{j} + 3\mathsf{c}$ gives the right side. Unit shortcuts, reused objects and an exact-root stop can only reduce the count. □

If charged, the final inversion adds 5j once, and supplying u_0 instead of s_0 another 5j.

5.2 Six supports at a common state

Table 5: Mobile work per step with the triangle power stage, integer $p \geq 3$.

Construction	Order	j	p	c	Traces	Scalar scope
MA rectangle	1	2	$2p+1$	0	$10p+7$	$X > 1$
AK rectangle	2	2	$2p+1$	0	$10p+7$	global
AD rectangle	3	3	$2p+1$	1	$10p+9$	global
Projective AD	4	4	$2p+1$	0	$10p+9$	global
Decentered AD	5	3	$2p+1$	1	$10p+9$	global, $p > 2$
Nested radicals	9	32	$2p+3$	12	$10p+59$	global, $p > 2$

For cube roots the per-step bounds are $37, 37, 39, 39, 39, 89$. Table 6 runs all six from the same state $X = 2$, $s_0 = 3/4$ (so $u_0 = 4/3$, not the $u_0 = 1$ of the illustration) with the same stopping test on the reciprocal relative error.

Table 6: Steps / summed trace bounds to reach $|s/\alpha - 1| \leq 10^{-d}$ for $p = 3$, $X = 2$, $s_0 = 3/4$. Each total is the per-step bound times the number of steps. Counts agree at 250 and 500 working digits.

Construction	Order	$d = 6$	$d = 12$	$d = 30$	$d = 60$
MA rectangle	1	8 / 296	17 / 629	44 / 1628	90 / 3330
AK rectangle	2	3 / 111	4 / 148	5 / 185	6 / 222
AD rectangle	3	2 / 78	3 / 117	3 / 117	4 / 156
Projective AD	4	2 / 78	2 / 78	3 / 117	3 / 117
Decentered AD	5	1 / 39	2 / 78	2 / 78	3 / 117
Nested radicals	9	1 / 89	1 / 89	2 / 178	2 / 178

For cube roots the nested support needs fewer steps but more traces than the decentered arc at every target in the table: $89, 89, 178, 178$ against $39, 78, 78, 117$. Higher order alone does not give a drawing-work advantage. The next subsection shows that this is a statement about small p.

5.3 Efficiency index and the crossover at $p = 13$

For an iteration of order ϱ costing N units per step, the classical efficiency index is $\varrho^{1/N}$ [10, 9]: the number of correct digits is asymptotically multiplied by $\varrho^{1/N}$ per unit of work. We compare its logarithm $\iota = \log \varrho / N(p)$, with $N(p)$ the trace bound of Table 5.

Proposition 5.2 (Efficiency ordering)**.** *For every integer $p \geq 3$, $\iota_{\mathrm{AK}} < \iota_{\mathrm{AD}} < \iota_{\mathrm{proj}} < \iota_{\mathrm{dec}}$. The nested support satisfies $\iota_{\mathrm{nest}} > \iota_{\mathrm{dec}}$ if and only if*

$$p > \frac{59 \log 5 - 9 \log 9}{10 \log(9/5)} = 12.79\ldots,$$

that is, for integer degrees, if and only if $p \geq 13$. It exceeds ι_{AD} from $p = 5$ and ι_{proj} from $p = 8$.

Proof. The last three of the first group share $N = 10p + 9$ and have increasing orders, and $\log 2/(10p + 7) < \log 3/(10p + 9)$ because $10p \log(3/2) > 9 \log 2 - 7 \log 3$, whose right side is negative. The inequality $\log 9/(10p + 59) > \log 5/(10p + 9)$ is linear in p and rearranges to the stated threshold; the comparisons with orders three and four give $p > 4.1$ and $p > 7.6$ in the same way. □

The reason is visible in Table 5: the power stage costs about $10p$ traces for every support, so the 50 additional traces of the nested support are a fixed overhead that large p amortizes, and the ratio of indices tends to $\log 9/\log 5 \approx 1.37$. The index is an asymptotic quantity built from upper inventories; at a finite accuracy the integer number of steps makes the comparison lumpy, as Table 8 shows.

At $p = 20$ the nested support has the smaller total at four of the five targets and loses at $d = 1000$, where saving a single step (five instead of six) does not pay for the higher step cost.

Table 7: $10^3\iota = 10^3 \log \varrho/N(p)$. The largest value in each column is in bold.

Construction	$p=3$	5	8	12	13	20	50
AK	18.73	12.16	7.97	5.46	5.06	3.35	1.37
AD	28.17	18.62	12.34	8.52	7.90	5.26	2.16
Projective AD	35.55	23.50	15.58	10.75	9.97	6.63	2.72
Decentered AD	**41.27**	**27.28**	**18.08**	**12.48**	11.58	7.70	3.16
Nested radicals	24.69	20.16	15.81	12.27	**11.63**	**8.48**	**3.93**

Table 8: Steps / summed trace bounds for $p = 20$, $X = 2$, $s_0 = 3/4$ (per-step bounds 209 and 259).

Construction	$d=30$	$d=60$	$d=250$	$d=1000$	$d=2500$
Decentered AD	4 / 836	5 / 1045	5 / 1045	6 / 1254	7 / 1463
Nested radicals	3 / 777	3 / 777	4 / 1036	5 / 1295	5 / 1295

6 Direct root enclosures for real degrees

The power stage can be avoided altogether by constructing the root directly. The input is a positive length X and a real degree p, also given as a length; the output is an approximation of $r = X^{1/p}$ with prescribed relative error. A real input length is assumed to be available: this does not assert that every real number is constructible from a unit segment. This calculator is the only one in the paper whose every step is a finite construction for irrational p. We first treat $X > 1$ and $p \geq 1$; the cases $X = 1$ and $p = 1$ are immediate, and Proposition 6.2 removes both restrictions.

6.1 One mean proportional per subdivision

Set $\theta = 1/p$, $a = 1$, $b = X$ and $\lambda = \theta$. Here a, b are moving rail values. The invariant is

$$0 < a < b, \qquad 0 \leq \lambda \leq 1, \qquad r = a^{1-\lambda} b^{\lambda}. \tag{6.1}$$

If $\lambda \in \{0, 1\}$ return the corresponding endpoint, which is exact. Otherwise construct $m = \sqrt{ab}$ with the classical mean-proportional semicircle [4] and replace

$$(a, b, \lambda) \longmapsto \begin{cases} (a,\ m,\ 2\lambda), & \lambda \leq \frac{1}{2}, \\ (m,\ b,\ 2\lambda - 1), & \lambda > \frac{1}{2}. \end{cases} \tag{6.2}$$

Both branches preserve (6.1). Thus p determines the choices and the weight, never the length of a power chain. After n subdivisions

$$\frac{b}{a} = X^{2^{-n}}, \qquad z := \frac{b}{a} - 1 \geq 0. \tag{6.3}$$

No logarithm, exponential or power s^p is evaluated. The branch test compares two constructed lengths, λ and $\frac{1}{2}$; for irrational p equality never occurs, and for dyadic $1/p$ the process stops with an exact endpoint.

6.2 A fifth-order final enclosure

The harmonic–arithmetic readout of the companion gallery [5], $L_2 = ab/[(1-\lambda)b + \lambda a]$ and $U_2 = (1-\lambda)a + \lambda b$, encloses r by the weighted arithmetic–geometric mean inequality, with

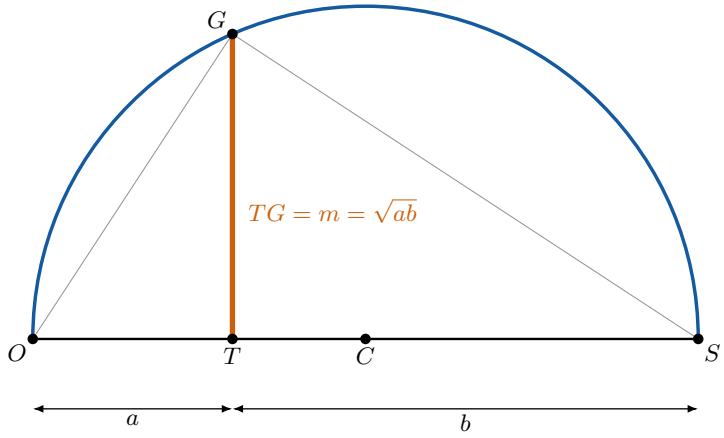
$$\frac{U_2}{L_2} - 1 = \lambda(1-\lambda)\frac{z^2}{1+z} \leq \sinh^2\Big(\frac{\log X}{2^{n+1}}\Big),$$

a width quadratic in z. A single rational readout of degree two gives a fifth-order width. It is the diagonal [2/2] Padé approximant of $(1+z)^{\tau}$ at zero [6, 7]. For $0 \leq \tau \leq 1$ put

$$\Pi_\tau(z) = 12 + 6(2+\tau)z + (\tau+1)(\tau+2)z^2, \qquad \psi_\tau(z) = \frac{\Pi_\tau(z)}{\Pi_{-\tau}(z)}, \tag{6.4}$$

whose denominators are positive for $z \geq 0$, and

$$L_5 = a\,\psi_\lambda(z), \qquad U_5 = \frac{b}{\psi_{1-\lambda}(z)}. \tag{6.5}$$



$\lambda \le \frac{1}{2}$: $(a, b, \lambda) \mapsto (a,\ m,\ 2\lambda)$

$\lambda > \frac{1}{2}$: $(a, b, \lambda) \mapsto (m,\ b,\ 2\lambda - 1)$

gap: $b/a \ \mapsto\ \sqrt{b/a}$

Figure 4: A subdivision constructs one mean proportional. The semicircle has diameter $OS = a + b$, centre C and junction $OT = a$; the orange perpendicular TG is the mean m. The same device is rotated onto the vertical rail in the counted protocol. The figure uses $a = 1.2$, $b = 2.8$ only to show the incidence.

Proposition 6.1 (Enclosure and finite termination)**.** *For $X \ge 1$, real $p \ge 1$ and every subdivision depth,*

$$L_5 \le X^{1/p} \le U_5, \qquad \frac{U_5}{L_5} - 1 = \frac{\lambda(1-\lambda)(2-\lambda)(1+\lambda)\, z^5}{\Pi_\lambda(z)\, \Pi_{1-\lambda}(z)} \le \frac{z^5}{256}. \tag{6.6}$$

Given $\varepsilon > 0$, choose a positive rational β with $\beta^5/256 \le \varepsilon$, stop when $b \le (1+\beta)a$ and return L_5. The procedure terminates and its relative error is at most ε. A sufficient depth depends on X and ε only, not on p.

Proof. For $0 \le \tau \le 1$, direct differentiation gives

$$(1+z)\big(\Pi'_\tau \Pi_{-\tau} - \Pi_\tau \Pi'_{-\tau}\big) - \tau\, \Pi_\tau \Pi_{-\tau} = -\tau(1-\tau^2)(4-\tau^2)\, z^4 \le 0.$$

Hence $\log \psi_\tau(z) - \tau \log(1+z)$ is nonincreasing from zero, so $\psi_\tau(z) \le (1+z)^\tau$. With $\tau = \lambda$ this is $L_5 \le r$; with $\tau = 1-\lambda$ and $(1+z)^\lambda (1+z)^{1-\lambda} = 1+z$ it is $r \le U_5$. Multiplying out gives the equality in (6.6). With $\chi = \lambda(1-\lambda) \le \frac{1}{4}$ the numerator coefficient is $\chi(2+\chi) \le \frac{9}{16}$, and $\Pi_\lambda, \Pi_{1-\lambda} \ge 12$, so the width is at most $z^5/256$. At the stopping test $0 \le z \le \beta$, so $r/L_5 - 1 \le \varepsilon$ and $(r - L_5)/r \le \varepsilon/(1+\varepsilon)$. By (6.3), $z \to 0$, so the test is eventually met; any n with $2^n \log(1+\beta) \ge \log X$ suffices. Equivalently $X \le (1+\beta)^{2^n}$, which can be checked in exact rational arithmetic for rational input. □

It suffices to draw L_5 alone; the analytic upper bound certifies its accuracy without constructing a second Padé quotient. For $0 < X < 1$, constructing and inverting this lower bound for $1/X$ gives an upper approximation of the original root with relative error at most ε. The width is of order five in the rail gap z; this is not a fifth-order iteration, since each subdivision only halves the logarithmic gap.

Proposition 6.2 (Every nonzero real degree and every positive input)**.** *Let $p \ne 0$ be real and $X > 0$, $X \ne 1$.*

(i) If $X > 1$ and $0 < p < 1$, let $k = \lfloor 1/p \rfloor \ge 1$ and start from $a = X^k$, $b = X^{k+1}$, $\lambda = 1/p - k$. Then (6.1), (6.3) and Proposition 6.1 hold unchanged.

(ii) If $0 < X < 1$, apply the construction to $1/X$ and invert: the enclosure $[1/U_5, 1/L_5]$ has the same relative width.

(iii) If $p < 0$, apply the construction to $|p|$ and invert the output.

Proof. In (i), $a^{1-\lambda} b^\lambda = X^{k+\lambda} = X^{1/p}$, $0 \le \lambda < 1$ and $b/a = X$, so the invariant and the gap formula are those of the case $p \ge 1$, and the proof of Proposition 6.1 uses nothing else. Inversion maps an enclosure $[L, U]$ to $[1/U, 1/L]$ with the same ratio. □

6.3 A complete, conservative drawing inventory

Use the extended vertical rail $R(q)$, its origin $R(0)$ and the prepared centres of Section 2.2; X and p are given on the rail. The macros are:

- *Sum or nonnegative difference* of two rail values: one compass arc on the rail.
- *Product or quotient*: five joins (Section 2.2); by a prepared constant, two joins.
- *Mean proportional*, $2\mathsf{j}+\mathsf{p}+3\mathsf{c}$: form $R(a+b)$ with one arc and $R((a+b)/2)$ with the two-join fixed scale; draw the semicircle through $R(0)$; draw the horizontal through $R(a)$ with one parallel macro; transfer the resulting length to the rail with one arc.

The updated weight uses the fixed scale 2λ and, on a right branch, one subtraction. A subdivision with its weight update costs at most $4\mathsf{j}+\mathsf{p}+4\mathsf{c}$, and a stopping test, which scales a by the prepared $1+\beta$, two joins. For the final readout use

$$L_5 = a\Big(1+\lambda\,\frac{6z(2+z)}{\Pi_{-\lambda}(z)}\Big), \qquad \Pi_{-\lambda}(z) = 12+6(2-\lambda)z+(1-\lambda)(2-\lambda)z^2. \tag{6.7}$$

One evaluation schedule uses seven general products, namely $(1-\lambda)(2-\lambda)$, z^2, $(2-\lambda)z$, $(1-\lambda)(2-\lambda)z^2$, $z(2+z)$, the weighted quotient and the final scaling by a; two divisions, including b/a; two fixed scalings by 6; and seven sums or differences. Its cost is at most $49\mathsf{j}+7\mathsf{c}$.

Table 9: Trace inventory of the direct enclosure for $X>1$, $p\ge 1$.

Operation	j	p	c
Initial weight $\lambda = 1/p$	5	0	0
n subdivisions and weight updates	$4n$	n	$4n$
At most $n+1$ stopping tests	$2n+2$	0	0
One final lower Padé readout	49	0	7
Total, parallels expanded	$8n+56$	0	$7n+7$

Thus $15n+63$ traces suffice for $X>1$, $p\ge 1$. The other cases of Proposition 6.2 add: $5k+1$ traces in case (i), for the k products $X^2,\dots,X^{k+1}$ and the difference $1/p-k$; two five-join inversions in case (ii), giving $15n+73$ for $0<X<1$, $p\ge 1$; one five-join inversion in case (iii). Finding k is a decision and is not charged. Constructing both endpoints L_5 and U_5 would require additional work and is not included in this single-output count.

6.4 Depth, precision and comparison with the iterations

Table 10: Sufficient depths at $X=2$, uniformly in real $p\ge 1$. The Padé depths use $\beta = 19/100,\ 3/40,\ 3/250,\ 3/1000$; the inequalities $\beta^5/256\le\varepsilon$ and $(1+\beta)^{2^n}\ge 2$ verify them in rational arithmetic.

Relative width target	Harmonic–arithmetic depth	Padé depth	Trace bound $15n+63$
10^{-6}	9	2	93
10^{-8}	12	4	123
10^{-12}	19	6	153
10^{-15}	24	8	183

For $p=37.5$, four subdivisions yield $a=1$, $b=2^{1/16}$ and $\lambda=32/75$, and the root lies between 1.0186558096 and 1.0186558103; these rounded endpoints can be checked exactly by raising them to the power 75 and comparing with 4. At $X=2$ and $\varepsilon=10^{-8}$ the construction has the bound $88\mathsf{j}+35\mathsf{c}=123$ traces.

When to use which calculator. The subdivision gains a fixed number of bits per step, the iterations multiply the number of correct digits. For cube roots of 2, the depths needed for width 10^{-d} with $d=6,12,30,60$ are $n=2,6,18,38$, that is $93,153,333,633$ traces, against $39,78,78,117$ for the decentered arc in Table 6. (The two columns are not the same quantity: a guaranteed width on one side, an observed error on the other.) For small integer or rational degrees and high accuracy the iterations are cheaper; for

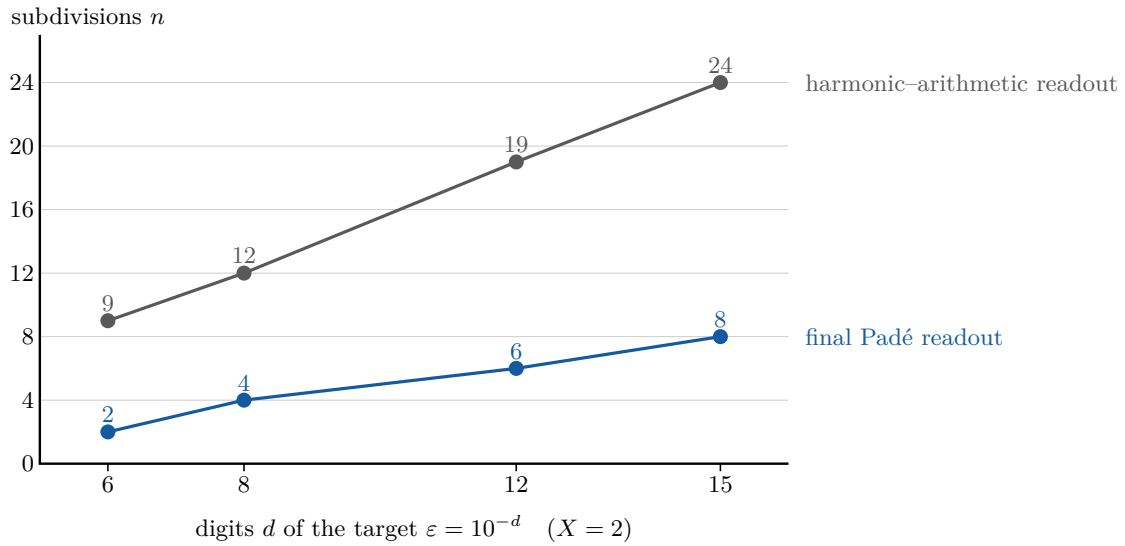


Figure 5: Sufficient worst-case depths of Table 10, uniformly in $p \geq 1$. Points are verified depth certificates; the connecting segments only guide the eye and do not represent fractional subdivisions.

irrational degrees, and for rational degrees with a large numerator, the subdivision is the calculator to use (Remark 4.8).

7 Attribution and algebraic approximation

High local order and geometric iteration are not new principles. Traub organizes classical iteration formulas and their error terms [8, 9]. Gander discusses Halley's method and radical corrections [11]; Scavo and Thoo analyze its geometry [12]. Melman connects geometry with global convergence, and Amat, Busquier and Gutiérrez develop geometric constructions for nonlinear iteration [13, 14]. Here AK is Newton under inversion and centered AD is Halley, realized in the shared rectangle; their scalar formulas are classical. Reading all supports as approximations of one function κ_* (Lemma 2.1) is the usual principle that the order of an iteration is the order of contact of its iteration function with the exact one, specialized to the rectangle.

The decentered and nested supports are algebraic corrections defined by explicit positive radicals. Algebraic approximants generalize rational approximation and form an established methodology [3, 15]; the two-radical coefficient matching of Section 4 belongs to it. Section 4.6 proves a degree-four distinction at the cubic specialization, and Appendix C gives a specified quadratic comparison for the decentered support. The subdivision mean proportional is Euclid's [4]; its final readout is an ordinary Padé approximant [6, 7], with the enclosure and width proved in Section 6. Counting particular constructions follows the concerns of géométrographie and nomography [16, 17], and the efficiency index is Ostrowski's [10].

What this paper contributes is the stated rectangle realizations, their domains, the scalar convergence proofs with explicit constants, the protocol bounds, and the classification of degrees.

8 Evidence and reproducibility

Lean. The nested-radical results have two project Docker/Lean audits totalling 104 theorems: 38 cubic, 14 real-parameter certificates, 44 real-parameter analysis and 8 degree-extension theorems. They use only `propext`, `Classical.choice` and `Quot.sound`, with no admitted proof. The sources, pinned toolchain, hashes and audit records accompany the paper [18, 19, 20]. The four nested-radical modules are those of V27-NR; for this revision they were recompiled from source in a separate environment (Lean 4.33.1, Mathlib v4.33.1, Mathlib itself built from source), the count of 104 theorems was confirmed, and the main theorems (global convergence, the extension to every nonzero degree and the cubic limit) were checked to depend on these three axioms only. This revision adds six modules, 125 theorems in all, with the same axioms and no admitted proof.

Table 11: Status of each claim. "Lean": checked by the Lean kernel, in the environments described below. "CAS": exact identity between polynomials or rational functions, recomputed with a computer-algebra system. "Numerical": high-precision evaluation, not a proof.

Claim	Domain	Status
Lemma 2.1, Table 3	analytic supports	Lean (limit form); constants of AK, AD and projective AD: CAS
Theorem 3.1	real $p > 2$	Lean (global convergence and the fifth-order limit); $O(e^7)$ remainder written
Theorem 4.1	real $p > 2$	Lean (limit $\hat{e}_+/\hat{e}^{\,9} \to c_9(p)$, positivity, $c_9(3)$); $O(\hat{e}^{\,11})$ remainder written
Theorem 4.3	$p = 3$, $X > 0$	Lean
Theorem 4.4	real $p > 2$	Lean; certificate structure also CAS
Proposition 4.5	rational p	Lean, for both radical supports
Proposition 4.6	real $p \neq 0$ (scalar)	Lean
Proposition 4.7	real $p > 0$	Lean
Proposition 5.1	integer $p \geq 3$	written inventory; totals Lean; incidence checks
Proposition 5.2, Table 7	real and integer $p \geq 3$	Lean; table values numerical
Tables 6, 8	stated examples	numerical
Proposition 6.1	$X \geq 1$, $p \geq 1$	written proof; scalar part in Lean (retained)
Proposition 6.2	real $p \neq 0$, $X > 0$	Lean (invariant, gap, inversion)
Degree four (Section 4.6)	$p = 3$, $t = 2$	exact rational computation

- `DecenteredArcGlobal` (30): Theorem 3.1, global monotone convergence of the decentered arc for every real $p > 2$, by the sign argument of Section 3.
- `ContactLimit` (12): Lemma 2.1 in limit form, and the analytic tool used for the local orders: a finite sum $\sum_j c_j e^{\lambda_j x}$ whose first N moments $\sum_j c_j \lambda_j^n$ vanish is $O(x^N)$.
- `NestedOrderNine` (21): Theorem 4.1, the limit $\hat{e}_+/\hat{e}^{\,9} \to c_9(p)$ for every real $p > 2$. The expansion (4.6) of Y through w^4 is an exponential sum of 27 terms with twelve vanishing moments; the nested radical is expanded algebraically and the result is identified with (4.9).
- `ArcOrderFive` (10): the fifth-order limit (3.2) with its constant.
- `DegreesEfficiency` (47), depending on Mathlib only: the exact contact identity; the positivity of N_9 and c_9, $c_9(3) = 7/648$ and $c_9(p) = \frac{1}{2}(p-1)p^7 D_4(1/p)$; Propositions 4.7 and 5.2, including $12 <$ crossover < 13 from $9^{129} < 5^{179}$ and $5^{189} < 9^{139}$; the trace totals of Section 5; the invariant of Proposition 6.2.
- `DegreesNested` (5): Proposition 4.5 for the nested support and for the decentered arc, without hypothesis and with integer powers only.

The local orders are stated as limits of $e_+/e^{\,\omega+1}$; the refinements $O(e^7)$ and $O(\hat{e}^{\,11})$ of the remainders, which follow from oddness, are written arguments.

Lean module	Certified scope
`NestedRadicalCubic`	Cubic global convergence and the relative-error limit 7/648, for every positive target.
`NestedRadicalRealCertificate`	Exact polynomial identities and positive-coefficient certificates.
`NestedRadicalReal`	Original formula for every real $p > 2$; support equivalence and the small-degree counterexample.
`NestedRadicalAllDegrees`	Degree doubling, preservation of the reported start and convergence for every $p \neq 0$.
`DecenteredArcGlobal`	New: global convergence of the decentered arc, real $p > 2$.
`ContactLimit`	New: contact lemma in limit form; exponential sums with vanishing moments.
`NestedOrderNine`	New: order nine with constant $c_9(p)$ for every real $p > 2$.
`ArcOrderFive`	New: order five of the decentered arc with its constant.
`DegreesEfficiency`	New: contact identity, constants, irrational degrees, efficiency ordering and crossover, trace totals, subdivision start for every positive degree.
`DegreesNested`	New: rational degrees for both radical iterations.
`GeometricSubdivision`	Scalar interval invariant, real-degree rational bracket and width; stored seven-declaration audit, not recompiled for this revision [21].

Computer algebra and numerics for this revision. The accompanying script `verify_v28.py` recomputes, with exact rational arithmetic: the identities (3.3) (at sample points), (3.4), (4.10); the series (4.6)–(4.9) and the constants of Table 3; the positivity of $N_9(2+\rho)$; the factorization defining $\mathcal{P}$ in Theorem 4.4, its degrees, and the positivity and term counts 345, 49, 24, 112 of the four certificates; the Padé identity and the width equality (6.6); and the appendix formulas. Numerically it checks (4.16) against a numerical derivative, the constant $c_9(p)$ against the iteration at $p = 2.5, 3, 4, 7, 20$, the monotone enclosure $\min(u, r) < F_{p,X}(u) < \max(u, r)$ on a grid $p \in [2.001, 10^4]$, $t \in [10^{-12}, 10^{12}]$, and regenerates Tables 6, 7, 8 and 10 (working precision 300 to 2600 digits).

Checks retained from V27. The nested verifier checks coefficient identities, the cubic positive certificate, the degree-four specialization, eight exact local series and 80 residual probes at 140 digits. The illustrative first step is checked at 110 digits without using the cube root to place points. The nested inventory has 64 exact rational macro checks and 60 full line/circle evaluations at 160 digits, including support placement and report; four additional root-stop cases are checked. The subdivision verifier checks three exact polynomial identities, 188 rational rail incidences, four rational depth certificates, the outward-rounded $p = 37.5$ example and 256 numerical enclosures at 100 digits [21]. The complete source package contains `build.sh`, `verify.py`, the independent comparison script and validation records. Finite numerical probes supplement proofs and do not certify rounded implementations. The gallery contains companion interactive views; it does not certify finite drawing error.

Figures. All figures are generated by `genfigs.py` from the formulas of the paper; the script asserts, for each drawing, that the constructed point P_+ agrees with the scalar update to 10^{-12}.

9 Scope and limits

The following limits apply to the whole paper and are stated here once.

1. **Approximation.** Every calculator produces a convergent sequence or a finite enclosure. No root is claimed to be exactly constructible in finitely many operations.
2. **Finite steps.** The fast iterations are finite constructions for integer and rational degrees only, at a cost governed by the numerator μ. For irrational p the convergence theorems are scalar statements; the geometric calculator is the subdivision.
3. **Counts.** Trace counts are upper bounds for the specified protocols under the convention of Section 2.4, which does not charge fixed preparation or decisions and weighs a join and an arc equally. They are not minimal, and they are not times, energies or drawing sizes. They are written inventories, not a formal cost semantics.
4. **Efficiency.** The index of Proposition 5.2 is asymptotic and inherits the previous limit; a shorter nested protocol would lower the crossover degree.
5. **Real drawings.** Exact incidences, the scalar real iteration and a physically inaccurate drawing are different objects. Conditioning, finite drawing error and floating-point behaviour are not analysed.
6. **Priority.** We have not identified the decentered or nested formulas with a prior formula, but a bounded comparison cannot establish priority, and different explicit formulas may be equivalent under changes of variable (Appendix C).
7. **Formalization.** Lean covers exactly the entries so marked in Table 11. The three convergence theorems and their local orders are formalized. Not formalized: the constants of the classical supports in Table 3, the higher-order remainders, the degree-four computation of Section 4.6 and Appendix C. For the inventories Lean checks the arithmetic of the totals, not the geometric protocols.

10 Conclusion and open questions

Changing one line in the Pandrosion rectangle moves the iteration from order two to order nine, and the order is read off the contact of that line's slope with a single function κ_*. Two radical supports extend the

classical ones: a decentered arc of order five at the mobile cost of Halley's construction, and a nested-radical support of order exactly nine for every real $p > 2$, with constant $c_9(p)$, globally monotone. In trace counts the decentered arc is the most efficient support for $p \le 12$ and the nested support for $p \ge 13$. These iterations are finite constructions for rational degrees; for a general real degree the power-free subdivision with a Padé readout gives a certified enclosure of relative width at most $z^5/256$ in $15n + 63$ traces.

Open questions:

1. Formalize a cost semantics for the drawing protocols, so that the inventories, and not only their totals, are machine-checked.
2. Shorten the nested inventory. Its overhead over the decentered arc is 50 traces, of which 19 build the two arguments K and ρ_9 and 20 the two mean proportionals; any saving lowers the crossover degree 13.
3. Find supports of contact five to seven. Is there a single-radical support, with one mean proportional, of order greater than five?
4. Analyse the two-level recursion in which the residual u^p of an irrational degree is itself enclosed by subdivision: which inner accuracy preserves the outer order?
5. Analyse conditioning and finite drawing error.
6. Compare the exact formulas with earlier algebraic approximants under specified transformations.

A Classical rectangle supports

A.1 AK and centered AD: orders two and three

AK: a fixed line. Prepare $K = (W(1 - X/p), 0)$. The support AK has slope $Hp/(WX)$, so $\kappa = p$ and $\mathrm{T}_{\mathrm{AK}}(s) = ps/(p - 1 + t)$, Newton's method under inversion. It has order two and converges from every positive start.

AD: a centered length transfer. Prepare

$$O_{\mathrm{AD}} = \Big(W - \frac{2W}{p-1},\ H - \frac{H(p+1)}{X(p-1)}\Big).$$

Transfer $AE = Hs^p$ downward from this centre to D_{AD} and join AD_{AD}. Then $\kappa = \frac{1}{2}[p+1+(p-1)t]$, giving

$$\mathrm{T}_{\mathrm{AD}}(s) = s\,\frac{p+1+(p-1)t}{p-1+(p+1)t}. \tag{A.1}$$

This Halley map has cubic order and global monotone convergence. Earlier geometric interpretations appear in [11, 12]; here the rail encoding and readout are shared with AK.

Both drawings retain the triangle stage, the green transport and the horizontal readout; only the support changes.

A.2 Projective AD: order four

Two joins supply the support. Prepare the horizontal line and the point

$$\ell:\ y = H\Big(1 + \frac{5p-1}{(p+1)X}\Big), \qquad Z_{\mathrm{proj}} = \Big(W\Big(1 + \frac{5p-1}{2p(2p-1)}\Big),\ H\Big(1 + \frac{p+1}{(2p-1)X}\Big)\Big).$$

(The line ℓ of this appendix is unrelated to the length ℓ of Figure 2.) Set $D_{\mathrm{proj}} = Z_{\mathrm{proj}}E \cap \ell$ and join AD_{proj}. Substitution in (2.3) gives the κ of Table 3 and

$$\mathrm{T}_{\mathrm{proj}}(s) = s\,\frac{2p\big((2p-1)t + p + 1\big)}{Q_{\mathrm{proj}}(t)}, \qquad Q_{\mathrm{proj}}(t) = (p+1)t^2 + 2(2p-1)(p+1)t + (2p-1)(p-1). \tag{A.2}$$

The direct-root correction is the [2/1] Padé approximant; the reciprocal-state factor displayed here is [1/2]. Direct differentiation gives

$$\mathrm{T}'_{\mathrm{proj}}(s) = -\frac{2p(p^2-1)(2p-1)(t-1)^3}{Q_{\mathrm{proj}}(t)^2},$$

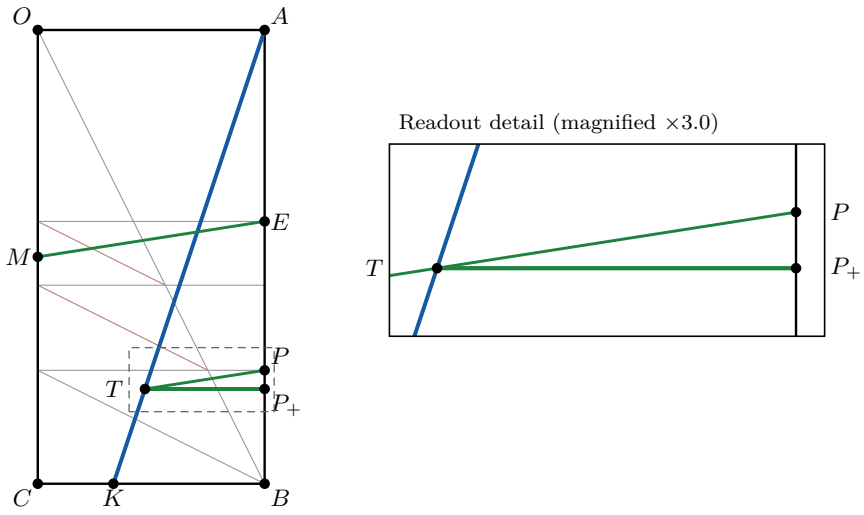


Figure 6: Pandrosion AK, order two, at $p = 3$, $X = 2$, $s = 3/4$ ($W = 2$, $H = 4$). The fixed support AK (blue) replaces the MA support, while the triangle chain (grey) and the correction direction (green) are retained. The inset magnifies the dashed window around P, T and P_+.

so the map has its global maximum α at $s = \alpha$. Also $\mathrm{T}_{\mathrm{proj}}(s)/s - 1 = (1-t)\big((p+1)t + 5p - 1\big)/Q_{\mathrm{proj}}(t)$. Thus a first step lands at or below α and subsequent steps increase to α; the order is exactly four. The earlier preprint gives further dynamics and the original triangle-chain realization [22].

B The linear MA reference

MA is a baseline, not an accelerated method. Its support is MB; write its lower-right endpoint as $A_{\mathrm{lin}} = B$. The parallel to ME through P meets MB, and a horizontal returns the next state. For $s \neq 1$,

$$\mathrm{T}_{\mathrm{MA}}(s) = 1 - \frac{X-1}{X\,\Sigma_p(s)}, \qquad \Sigma_p(s) = \sum_{j=0}^{p-1} s^j.$$

For $X > 1$ this map is increasing and $\mathrm{T}_{\mathrm{MA}}(s) - s = (1 - Xs^p)/(X\Sigma_p(s))$, so it converges monotonically to α from every positive start, with the removable extension at $s = 1$. Its derivative at α is $1 - p\alpha^{p-1}/\Sigma_p(\alpha) \in (0, 1)$: convergence is linear.

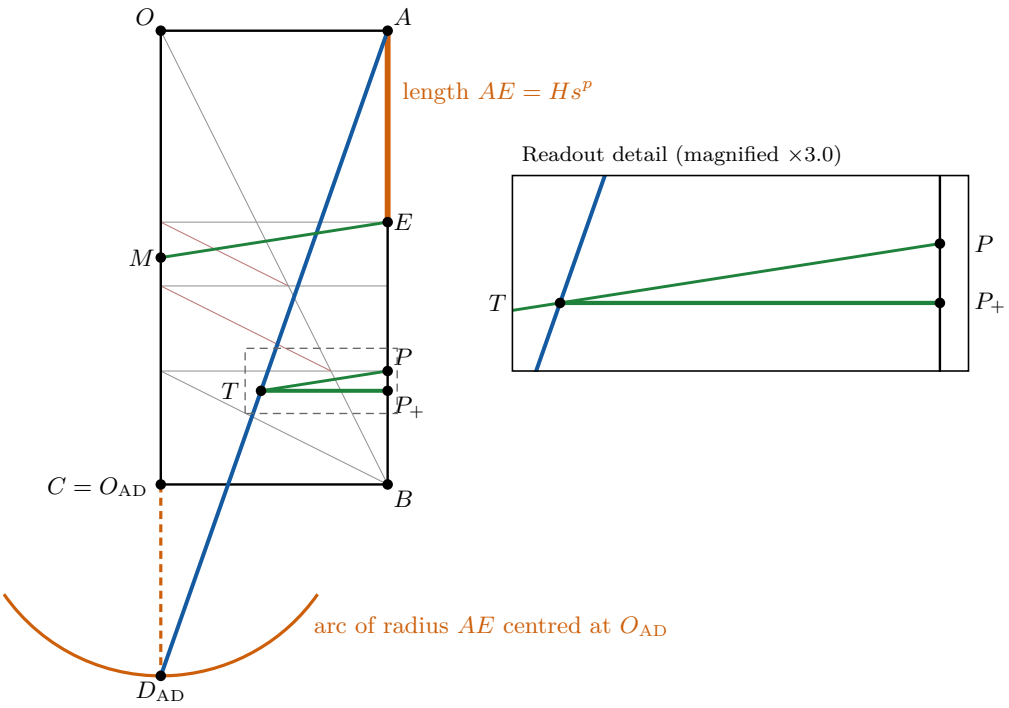


Figure 7: Pandrosion AD, order three, at the same state. The orange length AE is transferred onto the prepared vertical through O_{AD}; the lower arc point determines the support AD_{AD} (blue). Here $O_{AD} = C$.

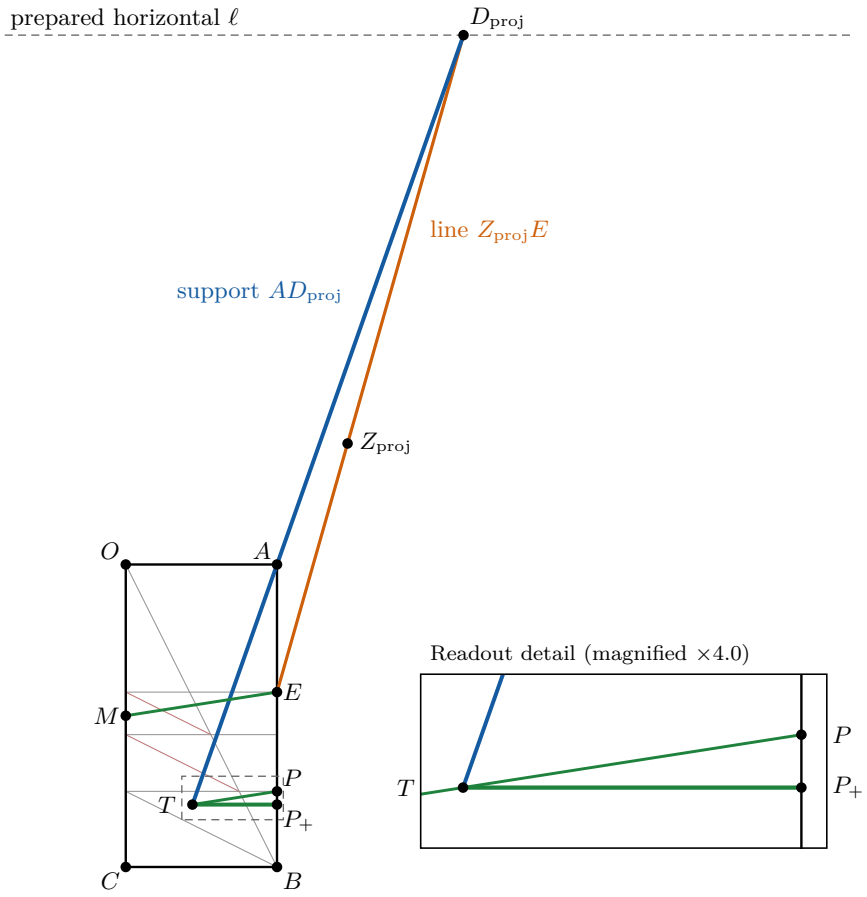


Figure 8: Projective AD, order four, at the same cubic state. The orange line $Z_{proj}E$ meets the prepared horizontal ℓ at D_{proj}; the blue support AD_{proj}, extended below A, receives the green correction. The inset magnifies the dashed window and separates P, T and P_+.

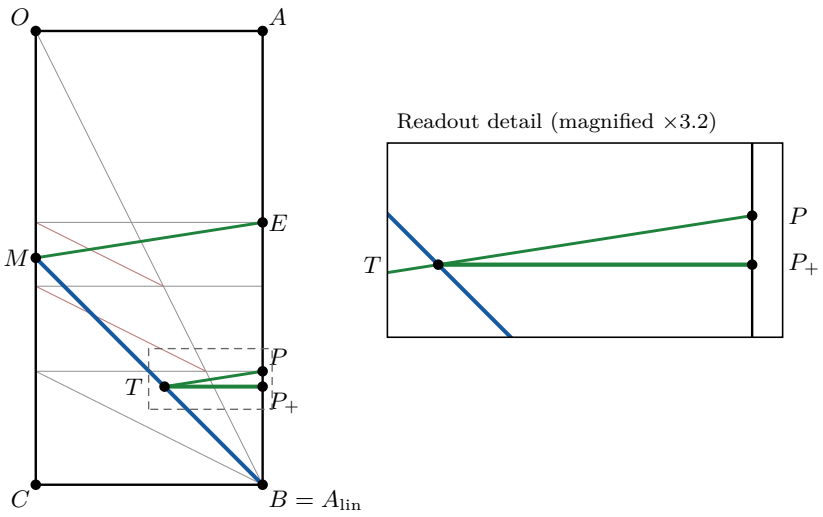


Figure 9: Pandrosion MA, the linear reference construction, at $p = 3$, $X = 2$, $s = 3/4$. Thin lines construct the power by similar triangles. The green parallel meets the blue support MA_{lin}; the horizontal gives P_+. The historical endpoint A_{lin} is the lower-right vertex B in this paper. At $s = 1$ the geometric intersection is coincident despite the removable scalar extension.

C Equivalence questions for the decentered correction

Whether the arc is a known formula "up to conjugacy" depends on the class of coordinate changes allowed. Let φ denote the correction factor $\Phi_{\mathrm{arc}}(t) = (1+g)/(t+g)$ with $g = (p-1)q_5$. Eliminating the radical gives the quadratic relation

$$P_p(t,\varphi) := (p-1)^2 S(t)(\varphi-1)^2 - 12p\,(t\varphi-1)^2 = 0. \tag{C.1}$$

The explicit radical in (3.1) selects the branch; the polynomial alone loses sign information and has a double root at $t = 1$.

Homographic changes. For each real $p > 2$, Φ_{arc} is not rational in t. Indeed $\mathrm{disc}_t\, S = 96p(p+1)$ and $S(0) = p-2 > 0$, so S has two simple nonzero roots and is not a square in $\mathbb{C}(t)$; if the correction were rational, so would be $\sqrt{S(t)/(12p)} = (1 - t\Phi_{\mathrm{arc}})/[(p-1)(\Phi_{\mathrm{arc}} - 1)]$. Invertible homographies of the input or output preserve rationality. For integer $p \geq 3$, $S(y^p)$ also has only simple roots, so the state map $y\,\Phi_{\mathrm{arc}}(y^p)$ is not rational and cannot be Möbius-conjugate to a rational state map, even as a germ at $y = 1$. This state-map exclusion is asserted only for integer degrees.

A specified quadratic normalization. A diagonal type $(2,2,2)$ quadratic Hermite–Padé (Shafer) form for $f(t) = t^{-1/p}$ requires polynomials of degree at most two with $A(t)f^2 + B(t)f + C(t) = O((t-1)^8)$ [23, eqs. (1.4)–(1.6)]. Direct substitution into (C.1) gives instead, with $\xi = t - 1$,

$$P_p\big(1+\xi, (1+\xi)^{-1/p}\big) = \frac{(p-2)(p-1)^2(p+1)(3p+1)}{60p^4}\,\xi^6 + O(\xi^7),$$

contact six, not eight. The coefficients of P_p have no common factor and its discriminant in φ is $48p(p-1)^2(t-1)^2 S(t)$, so it is irreducible over $\mathbb{C}(t)$; any polynomial with the same degree bounds defining this branch is a constant multiple of P_p. This excludes that particular diagonal approximant in these coordinates, without excluding other algebraic families or transformed targets.

Local analytic conjugacy. Distinct formulas may share a local analytic normal form: Böttcher's theorem conjugates a superattracting germ $F(e) = ce^{\varrho} + O(e^{\varrho+1})$, $c \neq 0$, to its monomial normal form [24]. This supplies neither a finite incidence protocol nor a global domain or drawing inventory. The explicit construction and its proved dynamics are therefore distinct from a claim of a new local conjugacy class.

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